Strategic Analysis of Card-Based Liar's Poker: Combinatorial Foundations and Nash Equilibrium Approximation

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April 7, 2026

Abstract

We study a card-based variant of Liar's Poker played with a standard 52-card deck, where players bid on the quality of the combined pool of all players' hidden hands using standard poker hand rankings. The game proceeds through a series of escalating bids until a player calls "bullshit," triggering a pool-wide reveal. A player is eliminated upon accumulating five cards. This paper establishes the combinatorial foundations for rational play by computing exact hand-type probabilities for pool sizes $n = 5$ through $n = 25$ cards, enabling provably optimal "blind" bidding strategies that require no private information. We further characterize how a player's private hand shifts the probability distribution over pool outcomes, quantifying the information value of observing one's own cards. These results provide the analytical groundwork for a forthcoming implementation of a Regularized Nash Dynamics (RND) agent [1] trained via self-play, using the OpenSpiel framework [2] as a computational backbone.

1 Introduction

Liar's Poker occupies a unique position in the landscape of imperfect-information games: it combines probabilistic reasoning over a shared hidden state with strategic bluffing and adversarial inference. The card-based variant studied here departs from the dollar-bill version popularized in finance by instead using poker hand rankings to define the bid space, creating a richer and more complex strategic environment.

This paper makes the following contributions:

1. We provide exact combinatorial probabilities for all ten standard poker hand types for pool sizes $n = 5$ to $n = 25$ cards drawn from a 52-card deck (Section 3.1), establishing a principled basis for blind bidding strategy.

2. We quantify the conditional shift in pool-hand probabilities given a player's private hand (Section 3.2), characterizing the value of private information.

3. We outline the game-theoretic framework and plan for a Regularized Nash Dynamics agent [1] to approximate the Nash equilibrium of the game (Section 4).

2 Game Rules and Formal Model

2.1 Setup and Play

The game is played by $k \in \{2, 3, 4, 5\}$ players using a standard 52-card deck. At the start of each round, every player is dealt exactly one card face-down from a freshly shuffled deck. Players

may look at their own card at any time but may never reveal it to other players.

Definition 1 (Hand Pool)**.** The *hand pool* is the multiset of all cards currently held by all active players. If player i holds h_i cards, the pool has size $n = \sum_i h_i$.

Definition 2 (Pool Hand)**.** The *pool hand* is the best standard 5-card poker hand achievable from any five cards selected from the hand pool. Hand types, from lowest to highest, are:

$$\text{High Card} < \text{Pair} < \text{Two Pair} < \text{Three of a Kind} < \text{Straight} <$$

$$\text{Flush} < \text{Full House} < \text{Four of a Kind} < \text{Straight Flush} < \text{Royal Flush}.$$

2.2 Bidding Protocol

Play proceeds in turns in a fixed circular order. On each turn, the active player must choose one of two actions:

1. **Bid:** Name a poker hand (T, r) where T is a hand type and r is a primary rank specifier. The bid asserts that the hand pool contains a 5-card combination of type T at rank r or better. The bid must be strictly higher than the previous bid, where bids are ordered first by hand type T, then by primary rank r within the same type.
2. **Call:** Challenge the previous bidder's claim. All cards in the hand pool are then revealed and the pool hand is evaluated.

Definition 3 (Bid Ordering)**.** Bid $(T_1, r_1) > (T_2, r_2)$ if and only if $T_1 > T_2$, or $T_1 = T_2$ and $r_1 > r_2$. The first bid of a round may be any valid bid.

2.3 Challenge Resolution and Elimination

When a player calls, the hand pool is revealed and the pool hand is evaluated.

- **Correct call** (the claimed hand does *not* exist in the pool): The player who made the false bid receives one additional card dealt from the deck.
- **Incorrect call** (the claimed hand *does* exist in the pool): The player who called receives one additional card dealt from the deck.

In both cases, the player who gains a card now holds one more card for subsequent rounds.

Definition 4 (Elimination)**.** A player is *eliminated* when they hold 5 or more cards. The last remaining player wins the game.

Remark 1. The symmetric penalty structure—both incorrect bids and incorrect challenges result in receiving an additional card—creates an asymmetric strategic landscape. Because cards accumulate toward the elimination threshold, *both* bluffing too aggressively (risking a correct challenge) and challenging too aggressively (risking an incorrect call) are penalized. This balance is central to the Nash equilibrium analysis.

Example 1 (Two-Player Round)**.** Players A and B each hold 2 cards. The pool has size $n = 4$. Player A bids "Pair of Kings." Player B calls. The four cards are revealed; suppose the pool contains $\{\clubsuit A, \heartsuit A, \diamondsuit 7, \spadesuit 3\}$. The pool hand is "Pair of Aces," so the bid was valid. B's call was incorrect: B receives a card and now holds 3 cards for the next round.

3 Combinatorial Probability Analysis

3.1 Blind Strategy: Marginal Hand Probabilities

A *blind strategy* is one that ignores the player's own cards and reasons solely from the pool size n. While not optimal in general, blind strategies provide principled lower bounds on rational bidding: if a hand type has probability $p < 0.5$ of appearing in a pool of size n, then bidding that type is disadvantageous without additional private information supporting the claim.

Definition 5 (Marginal Hand Probability)**.** For a pool of n cards drawn uniformly at random (without replacement) from a 52-card deck, the marginal probability of hand type T is

$$P(T, n) = \frac{|\{\, S \subseteq \binom{[52]}{n} : \text{best hand of } S = T \,\}|}{\binom{52}{n}}.$$

Figure 1 shows $P(T, n)$ for all ten hand types and pool sizes $n = 5$ through $n = 25$. For $n \in \{5, 6, 7\}$, values are exact combinatorial counts from published enumeration. For $n \geq 8$, values are computed via Monte Carlo simulation with $N = 3 \times 10^6$ samples per pool size; standard errors are below 0.01% for all reported values.

Wu and Wu [3] derive closed-form combinatorial formulas for the counts of n-card hands *containing* a straight, flush, or full house pattern for arbitrary n. Their exact counts serve as external validation of our Monte Carlo estimates and reveal two crossover points in the probability ordering of these hands: flush surpasses straight in occurrence frequency at $n = 12$, and full house surpasses straight at $n = 20$ (in the sense that more hands *contain* these patterns at those pool sizes). Note that Wu and Wu's definition counts any hand that contains the relevant pattern—including hands where a strictly better 5-card combination is also available—whereas our $P(T, n)$ records only the *best* achievable hand type. The two definitions coincide for $n \leq 7$ and diverge gradually for larger n as multi-pattern hands become common.

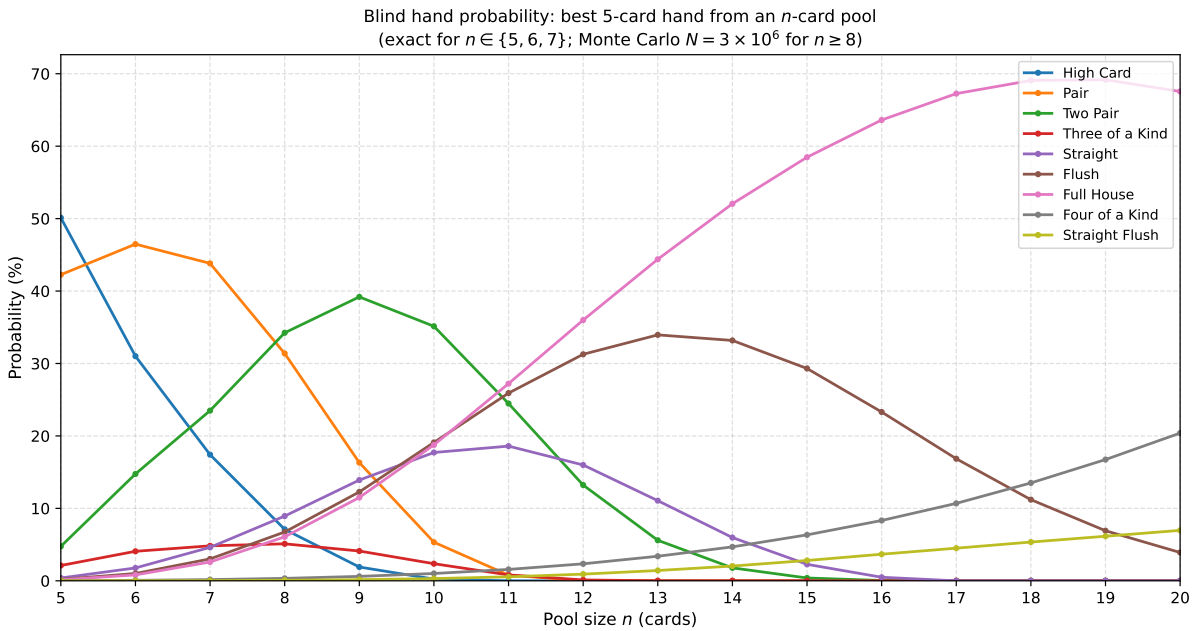

Figure 1: Probability of each best hand type as a function of pool size n (Royal Flush omitted; probability $< 3\%$ across the entire range). Generated by `generate_prob_tables.py`.

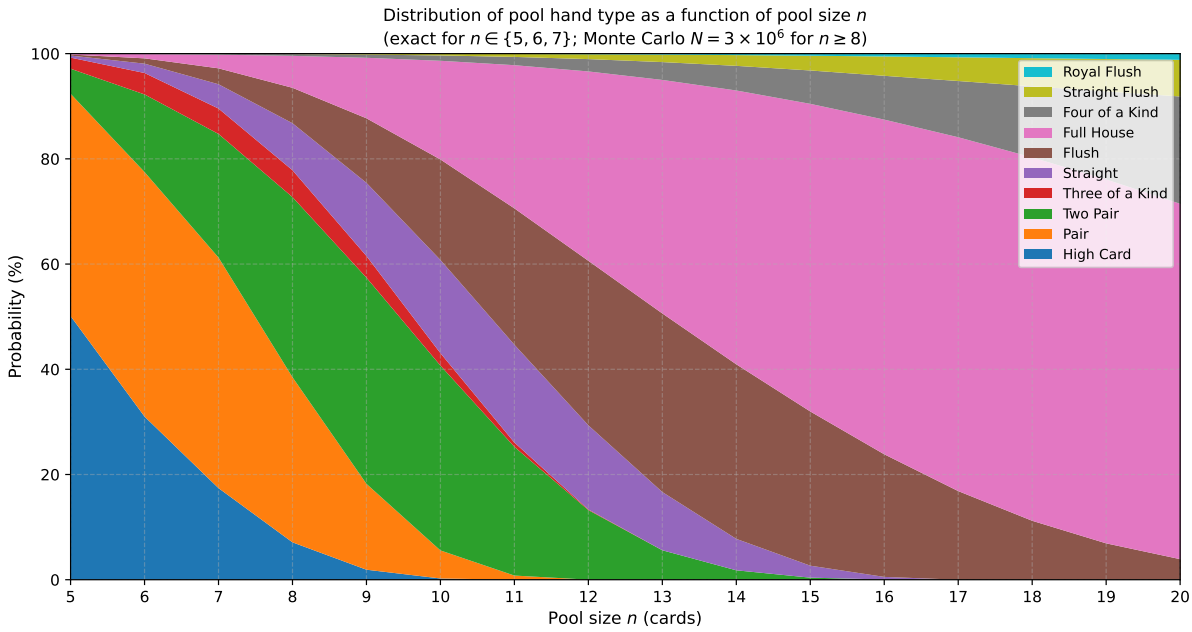

Figure 2: Stacked area chart showing how the full distribution of pool hand types shifts with pool size n. Each band gives $P(T, n)$; bands sum to 100%. Generated by `generate_prob_tables.py`.

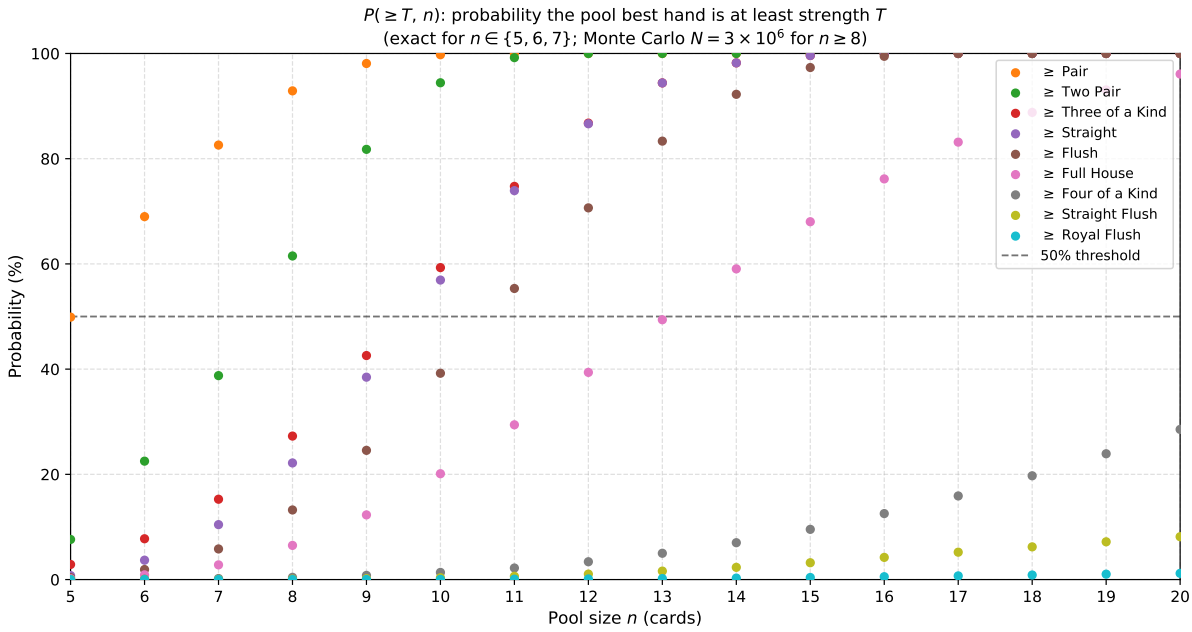

Figure 3: Cumulative "at-least" probability $P(\geq T, n) = \sum_{T' \geq T} P(T', n)$ for each hand type T as a function of pool size n. A value of 50% marks the threshold above which a blind bid on hand type T is better than chance. Generated by `generate_prob_tables.py`.

3.2 Conditional Probabilities Given Private Information

In practice, each player knows their own cards. This section quantifies how private hand knowledge shifts the probability distribution over pool hand types.

Definition 6 (Conditional Pool Probability)**.** Let $\mathbf{h}$ be the set of cards held by player i, achieving private hand type $X = \text{BestHand}(\mathbf{h})$. For a pool of total size n (including $|\mathbf{h}|$ of the player's own cards and $n - |\mathbf{h}|$ opponent cards drawn from the remaining $52 - |\mathbf{h}|$ cards), the conditional

pool probability is

$$P(T \mid X, n) = \Pr[\text{BestHand}(\mathbf{h} \cup \mathbf{o}) = T],$$

where $\mathbf{o}$ is drawn uniformly from $\binom{[52]\setminus\mathbf{h}}{n-|\mathbf{h}|}$.

The following subsections examine $P(T \mid \mathbf{h}, n)$ and the corresponding cumulative $P(\geq T \mid \mathbf{h}, n)$ for five private hand conditions. In each figure, dashed lines reproduce the blind baseline from Section 3.1 and solid lines show the conditional distribution. All conditional figures are generated via Monte Carlo with $N = 1{,}000{,}000$ samples (`compute_conditional_probs.py`).

3.2.1 Paired and Grouped Cards

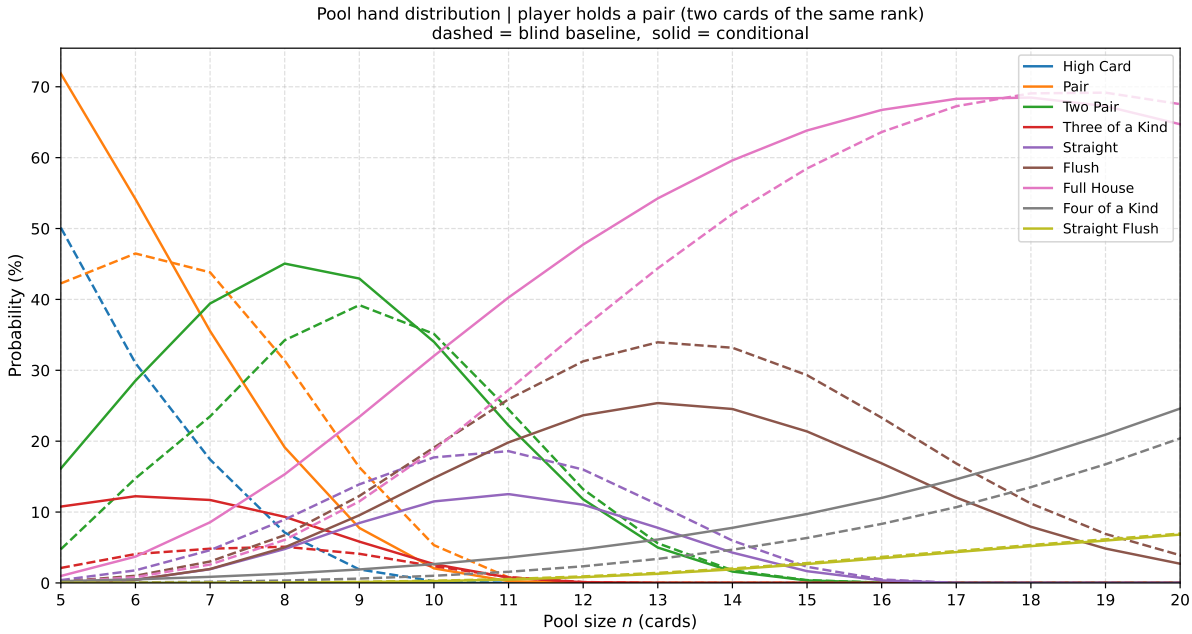

Figure 4: Conditional pool-hand distribution given the player holds a *pair* (two cards of the same rank). Dashed = blind baseline; solid = conditional.

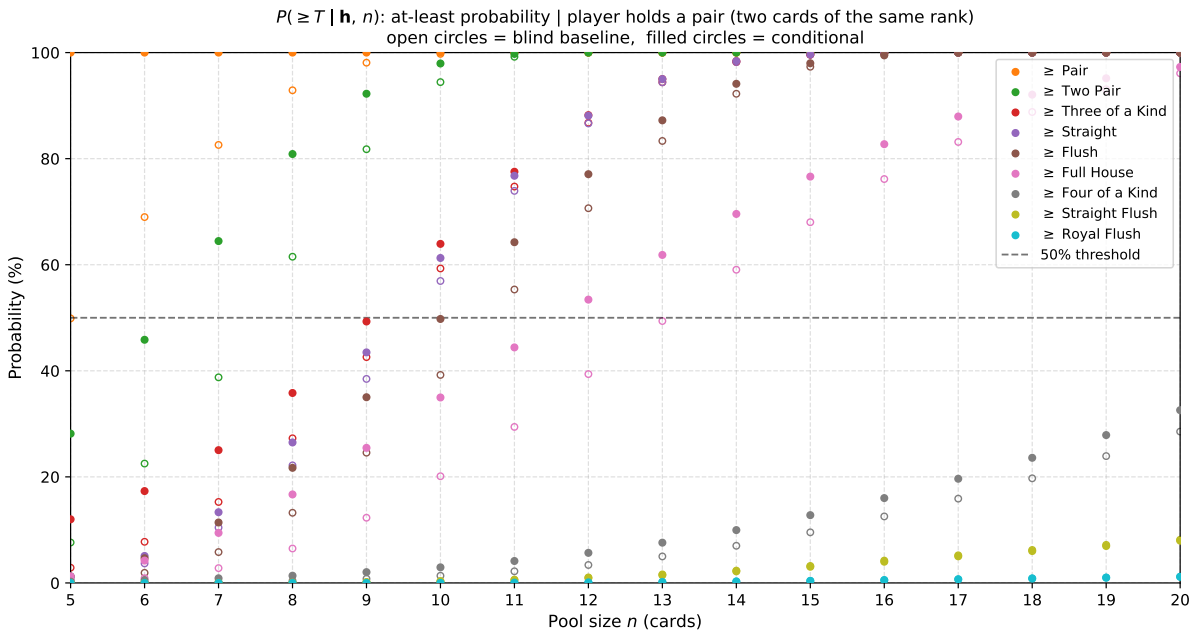

Figure 5: Cumulative at-least probability $P(\geq T \mid \mathbf{h}, n)$ given the player holds a *pair*. Open circles = blind baseline; filled circles = conditional; dashed = 50% threshold.

Holding a pair eliminates the possibility of a High Card pool outcome entirely across all pool sizes. At $n = 7$, the Two Pair probability rises by +15.9 pp to 39.4%, Three of a Kind by +6.9 pp to 11.7%, and Full House by +6.0 pp to 8.6% relative to the blind baseline; Straight and Flush are suppressed by 2.7 and 1.1 pp respectively. At $n = 10$, the Full House elevation reaches +13.3 pp (32.0% vs. 18.8%), while Straight and Flush are reduced by 6.2 and 4.3 pp. At $n = 14$,

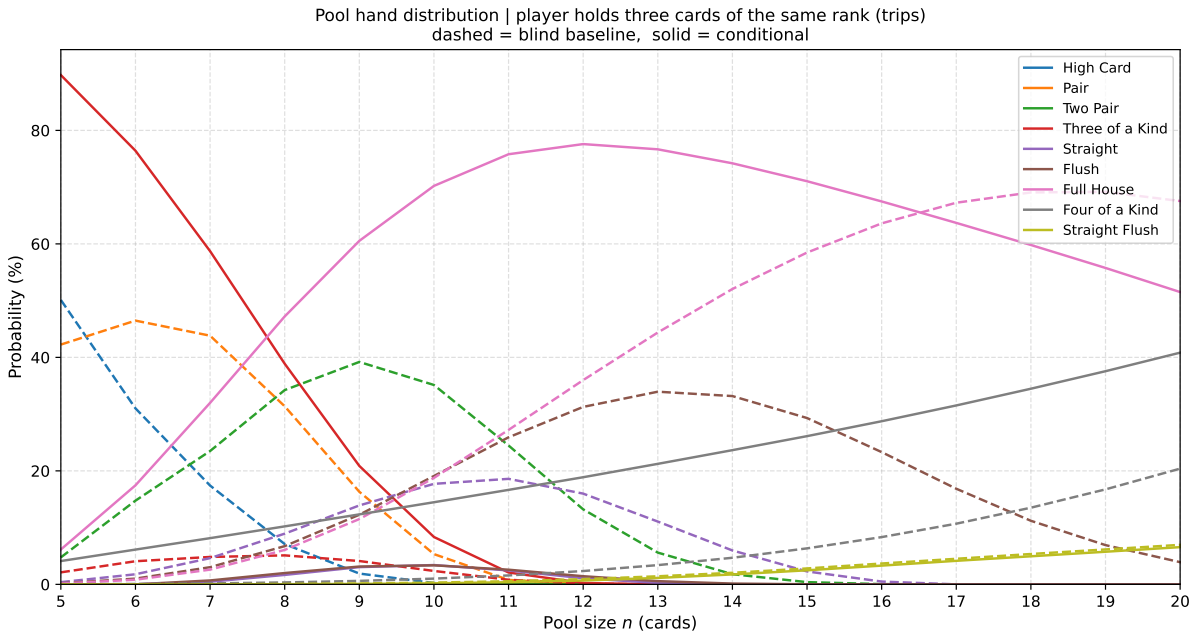

Figure 6: Conditional pool-hand distribution given the player holds *trips* (three cards of the same rank). Dashed = blind baseline; solid = conditional.

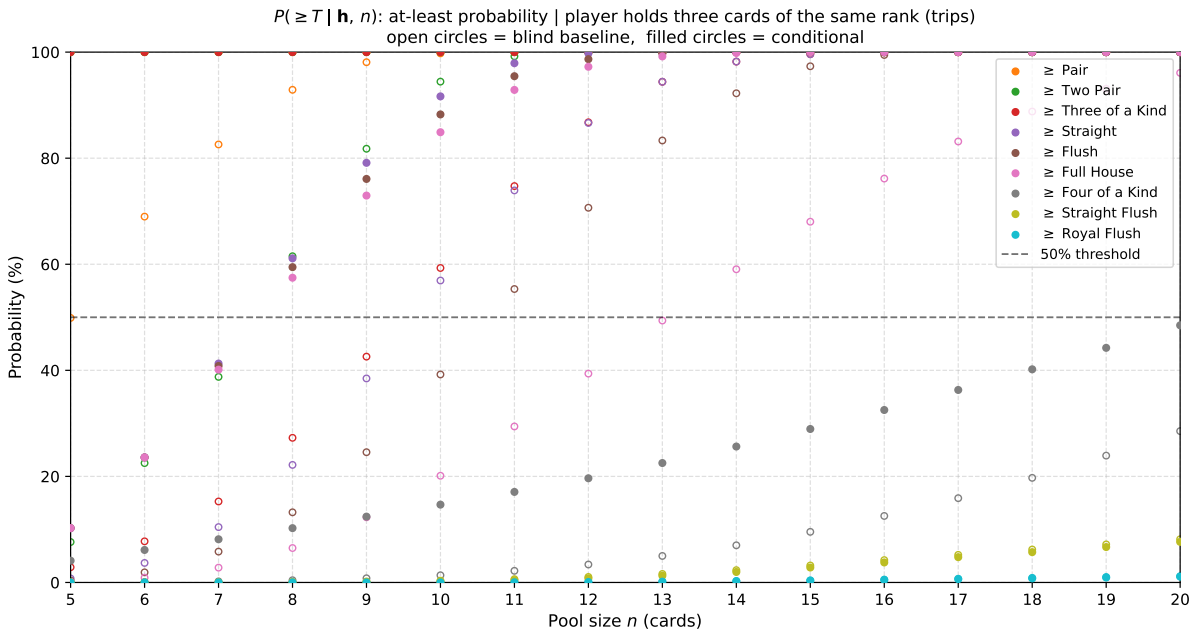

Figure 7: Cumulative at-least probability $P(\geq T \mid \mathbf{h}, n)$ given the player holds *trips*. Open circles = blind baseline; filled circles = conditional; dashed = 50% threshold.

the Flush suppression grows to −8.6 pp (24.5% vs. 33.2%) and the Full House premium narrows to +7.6 pp. These shifts support elevated bids on Full House and Four of a Kind relative to blind strategy and discount Straight and Flush bids at moderate pool sizes.

Holding trips eliminates High Card, Pair, and Two Pair as possible pool outcomes entirely. At $n = 7$, Three of a Kind rises by +53.9 pp to 58.7% and Full House by +29.4 pp to 32.0%, with Four of a Kind elevated to 8.1% (+8.0 pp). At $n = 10$, Full House dominates at 70.2% (+51.5 pp) and Four of a Kind rises to 14.5% (+13.5 pp); Straight and Flush are suppressed

by 14.3 and 15.7 pp respectively. At $n = 14$, Four of a Kind reaches 23.6% (+19.0 pp) and Full House 74.2% (+22.2 pp), while Flush is nearly entirely eliminated. At $n = 20$, Four of a Kind overtakes the relative distribution at 40.8% (+20.4 pp), an inversion relative to the baseline in which Full House remains the dominant outcome.

3.2.2 Suited Cards

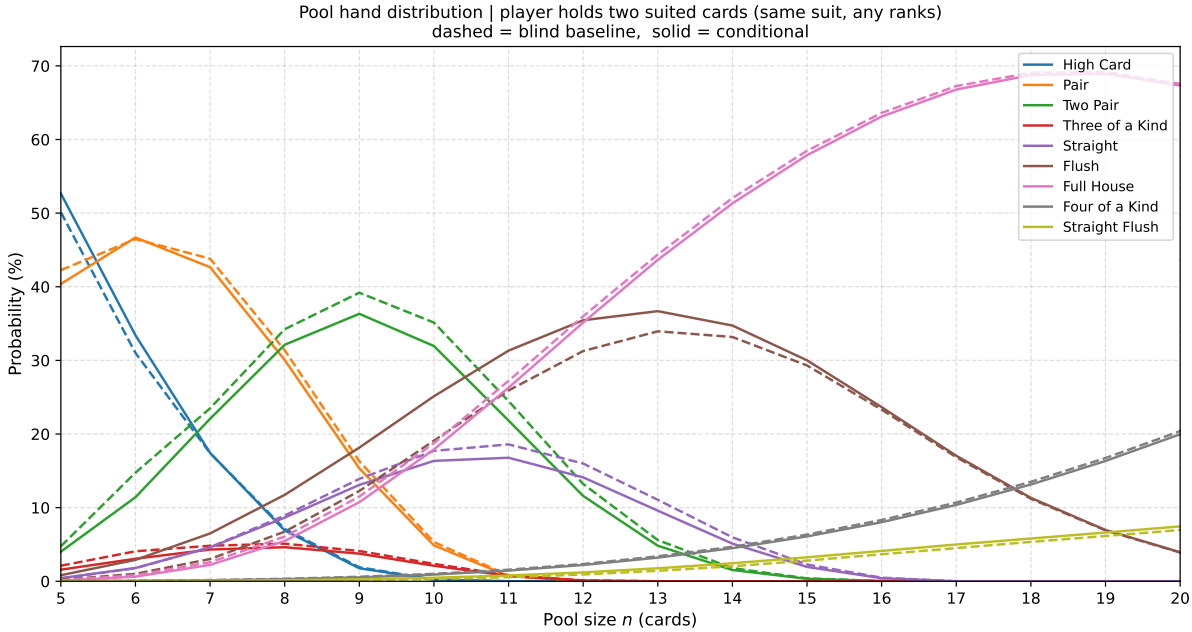

Figure 8: Conditional pool-hand distribution given the player holds two *suited cards* (same suit, any ranks). Dashed = blind baseline; solid = conditional.

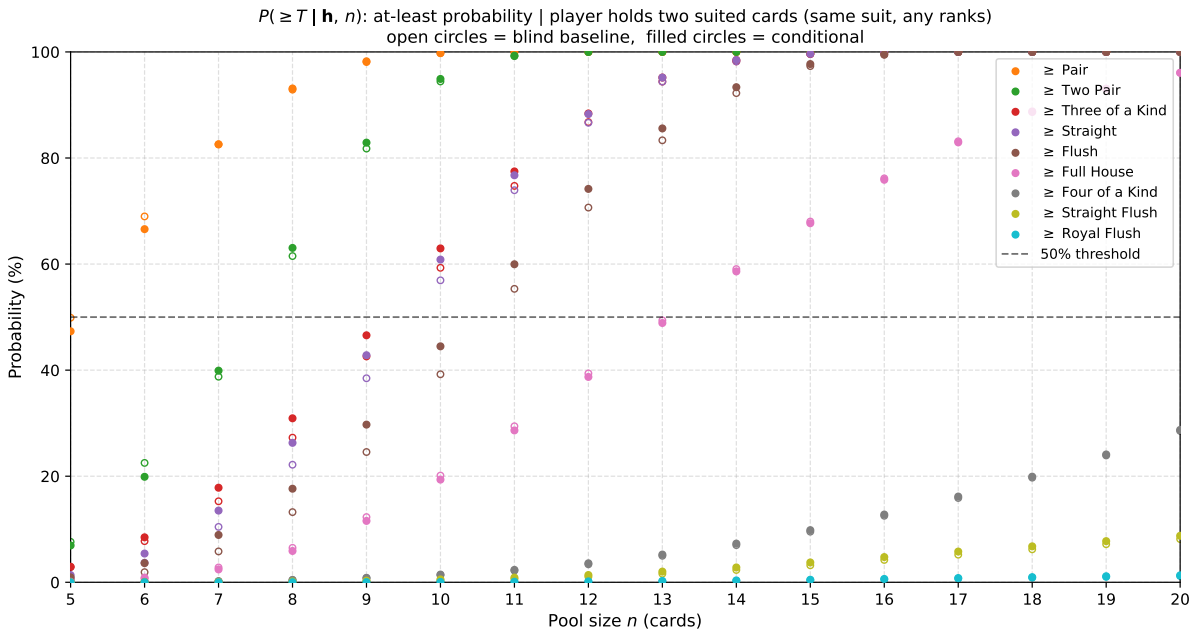

Figure 9: Cumulative at-least probability $P(\geq T \mid \mathbf{h}, n)$ given the player holds *two suited cards*. Open circles = blind baseline; filled circles = conditional; dashed = 50% threshold.

Holding two suited cards produces a moderate elevation in Flush probability. At $n = 7$, the Flush probability rises by +3.5 pp to 6.5%; at $n = 10$ by +6.0 pp to 25.1%. The effect narrows to +1.6 pp at $n = 14$ and is negligible at $n = 20$, reflecting the diminishing marginal contribution of two suit-matched private cards as the pool grows.

Holding three suited cards produces substantially larger Flush elevation. At $n = 7$, Flush

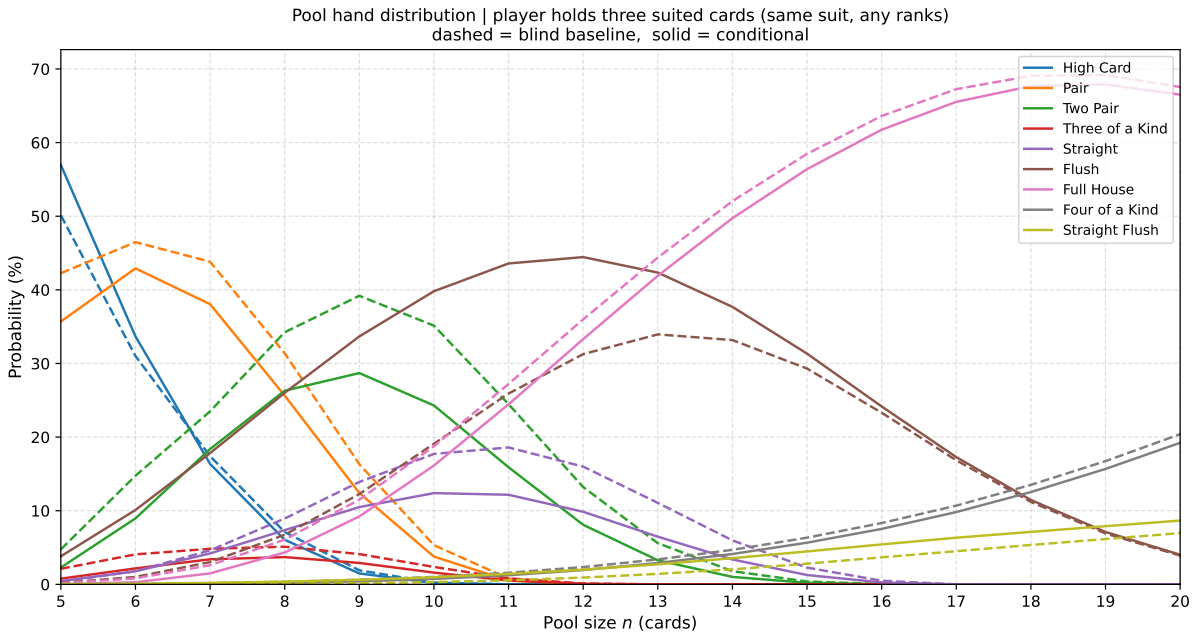

Figure 10: Conditional pool-hand distribution given the player holds *three suited cards* (same suit, any ranks). Dashed = blind baseline; solid = conditional.

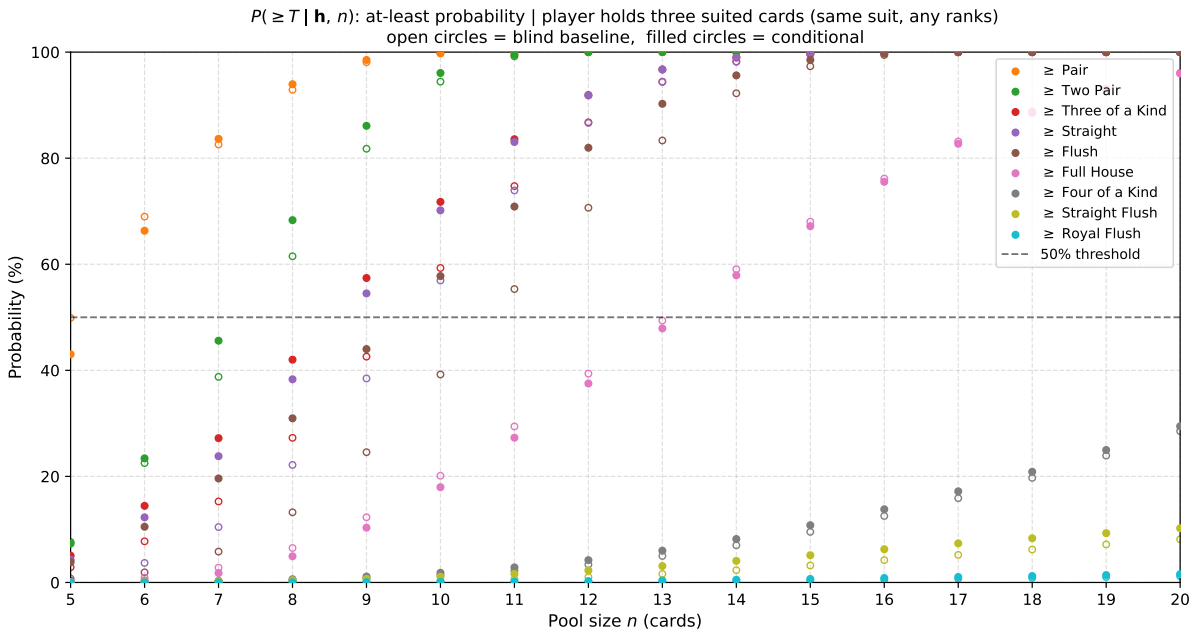

Figure 11: Cumulative at-least probability $P(\geq T \mid \mathbf{h}, n)$ given the player holds *three suited cards*. Open circles = blind baseline; filled circles = conditional; dashed = 50% threshold.

rises by +14.8 pp to 17.8% (approximately 4.2× the baseline rate), while Pair and Two Pair are each suppressed by roughly 5–6 pp. At $n = 10$, Flush reaches 39.8% (+20.7 pp), making it the single most probable pool outcome; Straight is reduced by 5.3 pp and Two Pair by 10.8 pp. At $n = 14$, the Flush elevation narrows to +4.5 pp (37.7% vs. 33.2%), and Straight Flush rises to 3.6% (+1.5 pp). At $n = 20$, a residual Straight Flush premium of +1.7 pp (8.7% vs. 7.0%) persists. These figures support Flush bids at small to moderate pool sizes and Straight Flush bids at larger pools when holding three suited private cards.

3.2.3 Adjacent and Range Cards

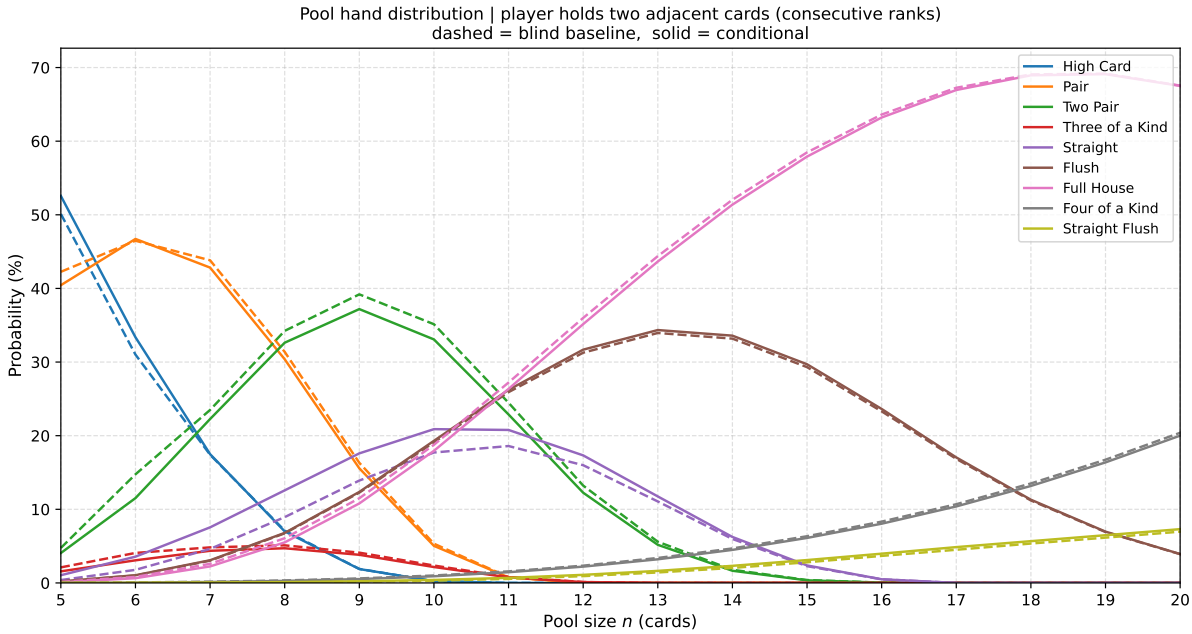

Figure 12: Conditional pool-hand distribution given the player holds two *adjacent cards* (consecutive ranks, any suits). Dashed = blind baseline; solid = conditional.

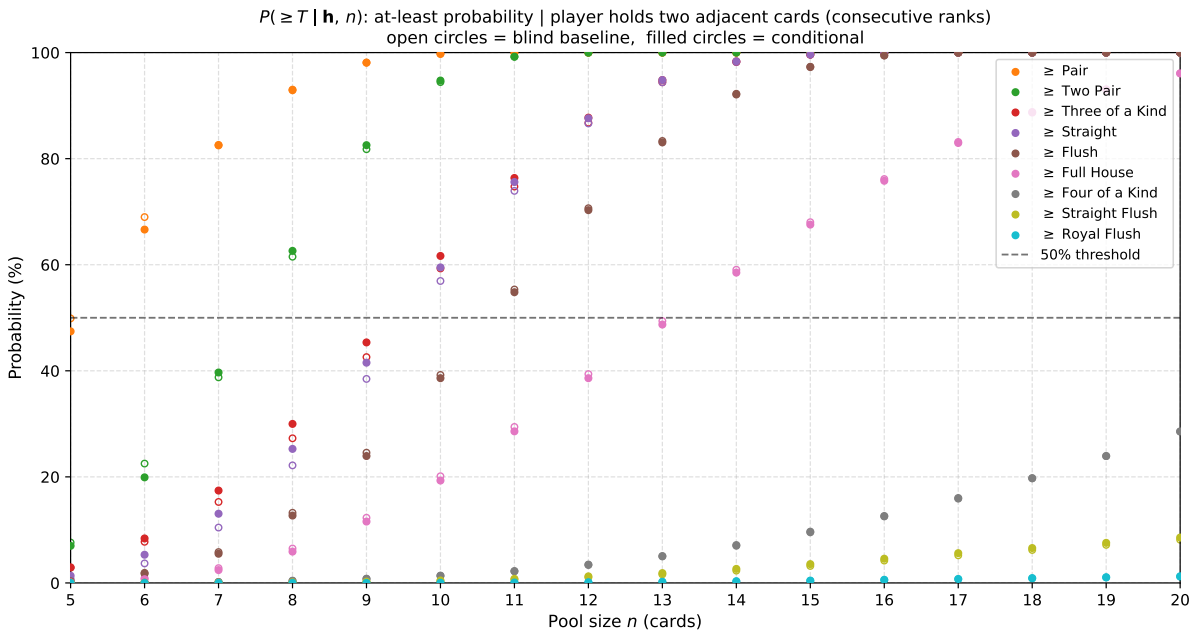

Figure 13: Cumulative at-least probability $P(\geq T \mid \mathbf{h}, n)$ given the player holds *two adjacent cards* (consecutive ranks). Open circles = blind baseline; filled circles = conditional; dashed = 50% threshold.

Holding two adjacent cards produces a modest but consistent Straight elevation. At $n = 7$, the Straight probability rises by +2.9 pp to 7.5%; at $n = 10$ by +3.2 pp to 20.9%. No significant deviations from baseline are observed at $n \geq 14$, indicating that the informational advantage of two consecutive private ranks is absorbed by the larger pool.

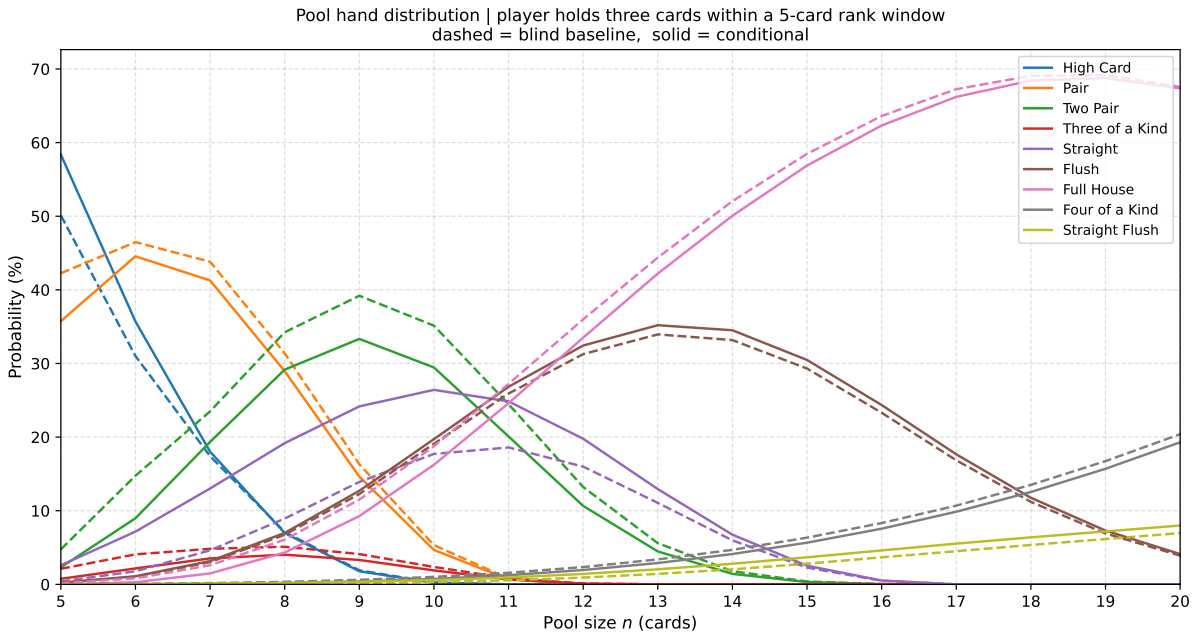

Figure 14: Conditional pool-hand distribution given the player holds *three cards within a 5-card rank window* (all ranks satisfy $\max - \min \leq 4$). Dashed = blind baseline; solid = conditional.

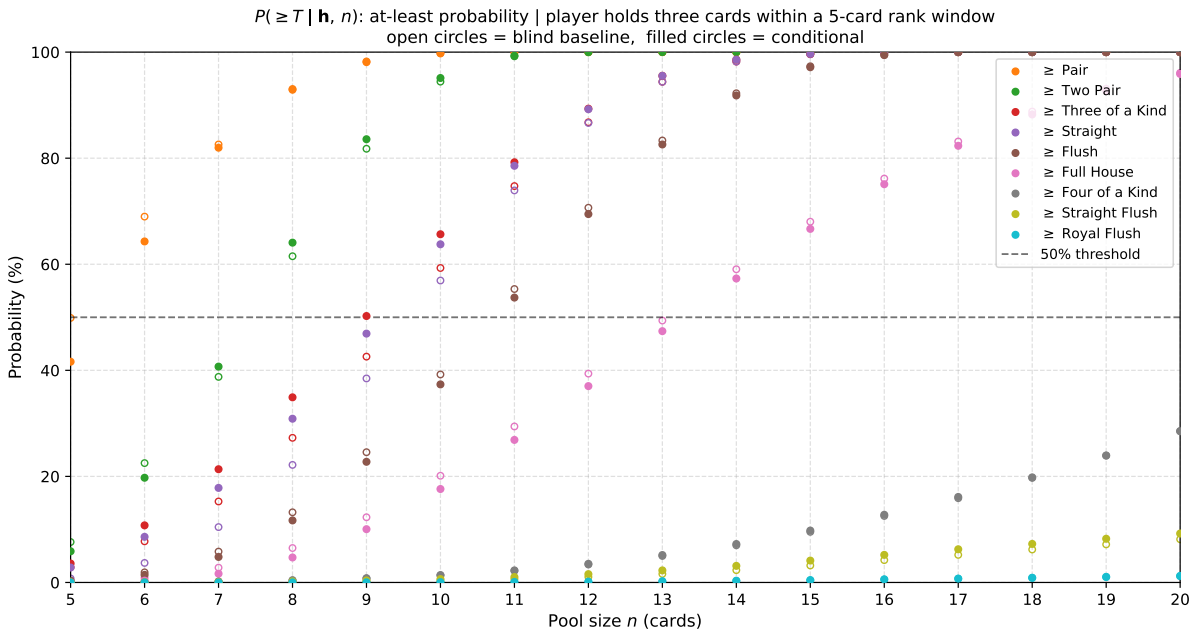

Figure 15: Cumulative at-least probability $P(\geq T \mid \mathbf{h}, n)$ given the player holds *three cards within a 5-card rank window*. Open circles = blind baseline; filled circles = conditional; dashed = 50% threshold.

Holding three cards within a 5-card rank window produces substantially larger Straight elevation. At $n = 7$, Straight rises by +8.4 pp to 13.0% (approximately 2.8× the baseline rate), while Two Pair is suppressed by 4.1 pp and Three of a Kind by 1.3 pp. At $n = 10$, the Straight elevation reaches +8.7 pp (26.4% vs. 17.7%), with Two Pair suppressed by 5.7 pp. The effect narrows sharply at $n = 14$ (Full House −2.0 pp; Flush +1.3 pp) and is negligible by $n = 20$, where only a +1.0 pp residual Straight Flush elevation persists.

3.2.4 Information Value of Private Cards

The analyses in the preceding subsubsections directly quantify the strategic value of private hand knowledge across six conditions. The paired and grouped conditions produce the largest rank-based deviations: a pair systematically elevates Full House and suppresses Flush and Straight, while trips eliminates lower hand categories entirely and drives Full House to 70% or greater at moderate pool sizes. The suited conditions produce the most pronounced Flush elevation, with three suited cards raising the Flush probability to nearly 40% at $n = 10$, a shift large enough to change the optimal bid type relative to blind strategy. The adjacent conditions produce the weakest deviations overall, with two adjacent cards providing only a 3 pp Straight premium that diminishes above $n = 14$. A formal treatment of information value via Kullback–Leibler divergence across all private-hand configurations is planned for subsequent work.

3.2.5 Rank-Level Bid Thresholds

The preceding analysis identifies which *hand type* straddles the 50% cumulative probability threshold for each condition and pool size. In practice, bids are made at the level of both hand type *and* primary rank: bidding "Pair" without specifying a rank is not a valid move. A more precise question is therefore: given pool size n and private-hand condition, what is the strongest bid (T, r) such that

$$P(\text{BestHand} \geq (T, r) \mid \mathbf{h},\, n) \;\geq\; 0.50\,?$$

We call this the *threshold bid*. Everything at or below the threshold bid is a "positive-expectation" bid (more likely true than false); everything above it is a "negative-expectation" bid.

For each condition and pool size we locate the pair $(T_{\text{above}}, r_{\text{above}})$ and $(T_{\text{below}}, r_{\text{below}})$ whose cumulative at-least probabilities bracket 50%, using the bid ordering of Definition 2: bids are ordered first by hand type, then by primary rank within a type. The primary rank encodes: the pair rank for Pair; the higher pair rank for Two Pair; the trip rank for Three of a Kind, Full House; the quad rank for Four of a Kind; the straight high card for Straight and Straight Flush; the flush high card for Flush; and always Ace for Royal Flush.

Tables **??** and **??** display these threshold bids for all seven scenarios (blind and six conditional conditions) across pool sizes $n = 5$ to 25. Each cell shows the bid label (e.g. **FH A** for Full House with trip rank Ace), its cumulative probability, and a canonical n-card example pool. Both tables are generated by `compute_conditional_probs.py` from 10^6 Monte Carlo samples per condition (3×10^6 for the blind scenario).

4 Future Work: Nash Equilibrium via Regularized Nash Dynamics

The combinatorial analysis of Sections 3.1 and 3.2 provides the information-theoretic foundation for strategic play. The next phase of this project is to approximate the Nash equilibrium of the game using Regularized Nash Dynamics (RND), a policy-gradient framework for imperfect-information games introduced in the context of Liar's Poker by [1].

4.1 Game-Theoretic Framework

We model the game as an extensive-form game with imperfect information. The information state of each player consists of:

- their own cards (observed),
- the sequence of bids made so far (public),
- the number of cards held by each player (public).

The action space at each information set is the set of valid bids (strictly higher than the current bid) plus the call action. The game is zero-sum in the sense that exactly one player wins.

4.2 Regularized Nash Dynamics

RND stabilizes the self-play training loop by adding an entropy regularization term that anchors the current policy toward a periodically updated reference policy. The update rule at training step t is:

$$\pi_{t+1} = \arg\max_{\pi}\; \mathbb{E}_{\pi}[G] - \eta \cdot \mathrm{KL}(\pi \,\|\, \bar{\pi}_t),$$

where G is the game value, η is a regularization coefficient, and $\bar{\pi}_t$ is the reference policy. The reference is updated every fixed number of steps to encourage exploration. The OpenSpiel framework [2] provides a principled computational environment for implementing and evaluating this training loop.

n	Blind		Pair		Trips		Suited	
5	Ace-High 69.3%	Pair of 2s 49.9%	Pair of Jacks 50.3%	Pair of Queens 44.7%	Three 9s 51.6%	Three 10s 44.7%	Ace-High 67.7%	Pair of 2s 47.3%
6	Pair of 6s 52.8%	Pair of 7s 49.1%	Pair of Aces 50.0%	2s Up 45.9%	Three 10s 52.9%	Three Jacks 47.1%	Pair of 6s 52.2%	Pair of 7s 48.6%
7	Pair of Jacks 52.4%	Pair of Queens 49.0%	10s Up 51.1%	Jacks Up 47.1%	Three Kings 50.3%	Three Aces 45.8%	Pair of Jacks 53.1%	Pair of Queens 49.8%
8	10s Up 50.3%	Jacks Up 46.9%	Kings Up 50.6%	Aces Up 43.6%	4s Full 50.5%	5s Full 47.0%	10s Up 52.6%	Jacks Up 49.3%
9	Kings Up 56.1%	Aces Up 49.8%	Aces Up 57.5%	Three 2s 49.3%	7s Full 50.8%	8s Full 46.2%	Aces Up 53.2%	Three 2s 46.6%
10	9-High Straight 50.4%	10-High Straight 48.8%	Broadway 51.5%	2-High Flush 49.8%	8s Full 54.5%	9s Full 49.0%	Queen-High Straight 50.2%	King-High Straight 48.7%
11	Queen-High Flush 50.9%	King-High Flush 46.7%	Ace-High Flush 52.5%	2s Full 44.4%	9s Full 55.1%	10s Full 49.1%	Queen-High Flush 54.7%	King-High Flush 49.8%
12	Ace-High Flush 52.3%	2s Full 39.4%	3s Full 50.3%	4s Full 47.0%	10s Full 53.6%	Jacks Full 47.3%	Ace-High Flush 53.5%	2s Full 38.7%
13	Ace-High Flush 63.6%	2s Full 49.4%	5s Full 51.4%	6s Full 47.7%	Jacks Full 51.2%	Queens Full 44.5%	Ace-High Flush 64.4%	2s Full 48.9%
14	4s Full 52.6%	5s Full 49.1%	7s Full 50.8%	8s Full 46.4%	Jacks Full 55.0%	Queens Full 48.2%	4s Full 52.2%	5s Full 48.8%
15	6s Full 53.8%	7s Full 49.8%	8s Full 53.1%	9s Full 48.3%	Queens Full 52.1%	Kings Full 44.9%	6s Full 53.6%	7s Full 49.6%
16	8s Full 53.1%	9s Full 48.4%	9s Full 54.9%	10s Full 49.5%	Queens Full 56.2%	Kings Full 49.0%	8s Full 53.0%	9s Full 48.3%
17	10s Full 50.2%	Jacks Full 44.4%	10s Full 55.9%	Jacks Full 49.8%	Kings Full 53.2%	Aces Full 45.1%	10s Full 50.2%	Jacks Full 44.4%
18	Jacks Full 51.0%	Queens Full 44.3%	Jacks Full 56.0%	Queens Full 49.1%	Kings Full 57.5%	Aces Full 49.4%	Jacks Full 51.0%	Queens Full 44.3%
19	Queens Full 50.6%	Kings Full 42.7%	Queens Full 55.1%	Kings Full 47.1%	Aces Full 53.7%	Four 2s 44.2%	Queens Full 50.6%	Kings Full 42.8%
20	Queens Full 57.0%	Kings Full 48.8%	Kings Full 52.8%	Aces Full 43.3%	Aces Full 58.1%	Four 2s 48.5%	Queens Full 57.1%	Kings Full 48.9%
21	Kings Full 55.0%	Aces Full 45.2%	Kings Full 58.7%	Aces Full 49.0%	Four 2s 53.0%	Four 3s 49.9%	Kings Full 55.1%	Aces Full 45.3%
22	Aces Full 51.3%	Four 2s 39.4%	Aces Full 54.9%	Four 2s 43.3%	Four 4s 51.2%	Four 5s 47.9%	Aces Full 51.4%	Four 2s 39.5%
23	Aces Full 57.6%	Four 2s 45.4%	Aces Full 60.9%	Four 2s 49.2%	Four 5s 52.4%	Four 6s 49.0%	Aces Full 57.7%	Four 2s 45.5%
24	Four 2s 51.8%	Four 3s 49.3%	Four 3s 52.6%	Four 4s 49.8%	Four 6s 53.5%	Four 7s 49.7%	Four 2s 51.9%	Four 3s 49.4%
25	Four 5s 50.1%	Four 6s 47.2%	Four 5s 52.8%	Four 6s 49.7%	Four 8s 50.3%	Four 9s 45.9%	Four 5s 50.2%	Four 6s 47.3%

Figure 16: Rank-level bid thresholds for the blind scenario and three private-hand conditions (Pair, Trips, Suited). Each cell is split left/right: the green left half shows the strongest bid (T, r) with $P(\geq (T, r)) \geq 50\%$, and the red right half shows the next stronger bid whose probability falls just below 50%. Generated by `compute_conditional_probs.py`.

n	3-Suited		Adjacent		3-Range	
5	Ace-High 65.0%	Pair of 2s 43.0%	Ace-High 65.1%	Pair of 2s 47.4%	Ace-High 56.8%	Pair of 2s 41.6%
6	Pair of 6s 53.1%	Pair of 7s 49.8%	Pair of 6s 52.6%	Pair of 7s 48.9%	Pair of 6s 53.1%	Pair of 7s 48.7%
7	Pair of Kings 51.5%	Pair of Aces 48.5%	Pair of Jacks 52.7%	Pair of Queens 49.3%	Pair of Jacks 51.6%	Pair of Queens 48.2%
8	Kings Up 50.4%	Aces Up 46.4%	10s Up 51.1%	Jacks Up 47.7%	10s Up 52.5%	Jacks Up 49.0%
9	9-High Straight 50.5%	10-High Straight 49.5%	Aces Up 51.2%	Three 2s 45.4%	Three 3s 50.1%	Three 4s 49.9%
10	Queen-High Flush 51.2%	King-High Flush 44.9%	10-High Straight 50.1%	Jack-High Straight 48.1%	10-High Straight 52.9%	Jack-High Straight 49.9%
11	King-High Flush 57.2%	Ace-High Flush 45.7%	Queen-High Flush 50.0%	King-High Flush 45.8%	Jack-High Flush 51.0%	Queen-High Flush 48.1%
12	Ace-High Flush 56.6%	2s Full 37.5%	Ace-High Flush 51.1%	2s Full 38.6%	King-High Flush 57.5%	Ace-High Flush 49.0%
13	Ace-High Flush 66.5%	2s Full 47.9%	Ace-High Flush 62.6%	2s Full 48.7%	Ace-High Flush 60.8%	2s Full 47.4%
14	4s Full 51.6%	5s Full 48.3%	4s Full 52.4%	5s Full 48.9%	4s Full 52.4%	5s Full 49.2%
15	6s Full 53.3%	7s Full 49.4%	6s Full 53.5%	7s Full 49.4%	6s Full 54.0%	7s Full 49.4%
16	8s Full 53.0%	9s Full 48.4%	8s Full 52.8%	9s Full 47.9%	8s Full 52.3%	9s Full 47.0%
17	10s Full 50.4%	Jacks Full 44.8%	9s Full 55.3%	10s Full 49.8%	9s Full 54.3%	10s Full 48.3%
18	Jacks Full 51.4%	Queens Full 44.8%	Jacks Full 50.5%	Queens Full 43.7%	10s Full 55.3%	Jacks Full 48.5%
19	Queens Full 51.1%	Kings Full 43.3%	Queens Full 50.0%	Kings Full 42.0%	Jacks Full 55.2%	Queens Full 47.8%
20	Queens Full 57.4%	Kings Full 49.3%	Queens Full 56.5%	Kings Full 48.1%	Queens Full 54.3%	Kings Full 46.1%
21	Kings Full 55.4%	Aces Full 45.7%	Kings Full 54.3%	Aces Full 44.4%	Kings Full 52.5%	Aces Full 43.3%
22	Aces Full 51.7%	Four 2s 39.9%	Aces Full 50.5%	Four 2s 39.3%	Kings Full 59.0%	Aces Full 49.5%
23	Aces Full 57.9%	Four 2s 45.9%	Aces Full 56.9%	Four 2s 45.4%	Aces Full 55.9%	Four 2s 45.2%
24	Four 2s 52.2%	Four 3s 49.7%	Four 2s 51.7%	Four 3s 49.5%	Four 2s 51.6%	Four 3s 49.6%
25	Four 5s 50.5%	Four 6s 47.7%	Four 5s 50.2%	Four 6s 47.3%	Four 5s 50.8%	Four 6s 47.9%

Figure 17: Rank-level bid thresholds for three additional private-hand conditions (3-Suited, Adjacent, 3-Range). Cell format as in Figure 16.

4.3 Implementation Plan

The implementation will proceed in three stages:

1. **Two-player, fixed pool size.** Solve the two-player game with fixed hand sizes (no card accumulation) to establish a baseline equilibrium.

2. **Two-player, full dynamics.** Add the card accumulation and elimination mechanic to study how the equilibrium shifts near the elimination boundary ($h_i = 4$).

3. **Multiplayer extension.** Generalize to $k \in \{3, 4, 5\}$ players, where coalition-like dynamics may emerge even in a nominally zero-sum setting.

The probability tables from Section 3 serve as oracle lookup tables within the agent's value network, reducing the computational burden of on-the-fly probability estimation during self-play.

5 Conclusion

This paper establishes the combinatorial foundations for rational play in card-based Liar's Poker. Exact and high-precision Monte Carlo probability tables for pool sizes $n = 5$ through $n = 25$ enable principled blind bidding strategies, and conditional probability analysis quantifies the strategic value of private card information. These results will directly inform the design of a Regularized Nash Dynamics agent in subsequent work, which aims to approximate the true Nash equilibrium of the game and provide strategic insights unavailable from purely combinatorial analysis.

References

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